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Flexible instrument delivery device

Abstract

A surgical access assembly may include an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween, and a flexible tool delivery device having a hub and a guide tube, the hub configured to selectively secure to a rim at the proximal end of the outer sheath, and the guide tube configured to extend at least partially into the outer sheath and provide a flexible tool proximate to a tissue.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application Ser. No. 63/219,085, filed Jul. 7, 2021, the disclosure of which is hereby incorporated in its entirety by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to a surgical access system for use with delicate and critical tissues, as well as methods of accessing and performing surgery using same. More specifically, the present disclosure relates to a flexible instrument delivery device for delivering tools, such as an imaging device to the tissues.

BACKGROUND

(2) Diagnosis and treatment of conditions affecting the human body, and specifically the brain, are among the most difficult and complex problems that face the medical profession. The brain is a complex and delicate soft multi-component tissue structure that controls bodily functions through a complex neural network connected to the rest of the body through the spinal cord. The brain and spinal cord are contained within and protected by significant bony structures, e.g., the skull and the spine. Given the difficulty of accessing the brain through the hard bony protective skull and the delicate network and complex interactions that form the neural communication network contained within the brain that define the human body's ability to carry on its functions of speech, sight, hearing, functional mobility, reasoning, emotions, respiration and other metabolic functions, the diagnosis and treatment of brain disorders presents unique challenges not encountered elsewhere in the body.

(3) For example, abnormalities such as intracranial cerebral hematomas (ICH), abscesses, glioblastomas (GB), metastases (mets) and functional diseases manifest themselves in the intraparenchymal subcortical space (i.e., the white matter) of the brain are particularly challenging to access, let alone treat. The ventricles of the brain contain eloquent communication structures (neural network) which are located in the subcortical space, called fiber tracts and fascicles. Thus, traditionally, unless the ICH, GB, and/or mets were considered anything but “superficial,” such conditions have been considered challenging to access, simply because getting to the abnormality ICH, GB and/or mets are considered just as damaging as letting the condition take its course. Similarly, tissue abnormalities such as tumors, cysts and fibrous membrane growths which manifest within the intraventricular space of the brain are considered challenging to safely access and often inoperable, due to their locations within the brain.

(4) In order to assist in diagnosis and subsequent treatment of tissue disorders, clear, accurate imaging of brain tissue through the skull is required. In recent years significant advances have been made in imaging technology, including stereotactic X-ray imaging, Computerized Axial Tomography (CAT), Computerized Tomographic Angiography (CTA), Position Emission Tomography (PET) and Magnetic Resonance Imaging (MM), Diffusion Tensor Imaging (DTI) and Navigation systems (instrument position tracking systems). These imaging devices and techniques permit the surgeon to observe conditions within the brain in a non-invasive manner without opening the skull, as well as provide a map of critical structures surrounding an area of interest, including structures such as blood vessels, membranes, tumor margins, cranial nerves, including fiber tracts and fascicles. If an abnormality is identified through the use of one or more imaging modalities and/or techniques, it may be necessary or desirable to biopsy or remove the abnormality.

SUMMARY

(5) A surgical access assembly may include an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween, and a flexible tool delivery device having a hub and a guide tube, the hub configured to selectively secure to a rim at the proximal end of the outer sheath, and the guide tube configured to extend at least partially into the outer sheath and provide a flexible tool proximate to a tissue.

(6) A flexible tool delivery device for a surgical system may include a hub configured to selectively

secure to a rim at a proximal end of an outer sheath of the surgical system, and a guide tube coupled to the hub and extending at least partially into the outer sheath to provide a flexible tool proximate to a tissue during use.

(7) A surgical access assembly may include an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween, and a flexible tool delivery device having a hub and a guide tube, the hub configured to selectively secure to a rim at the proximal end of the outer sheath, and the guide tube configured to extend at least partially into the outer sheath and provide a flexible tool proximate to a tissue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiments of the present disclosure will now be described in greater detail with reference to the attached figures, in which:
- (2) FIG. 1 illustrates a perspective view of a surgical access assembly having a flexible instrument arranged on a hollow outer sheath.
- (3) FIG. 2 illustrates a partial side view of the surgical access assembly of FIG. 1 with a guide tube in an uninstalled position.
- (4) FIG. 3 illustrates a partial side view of the surgical access assembly of FIG. 1 with a guide tube in an installed position.
- (5) FIG. 4 illustrates a rear perspective view of the flexible instrument of FIG. 1.
- (6) FIG. 5 illustrates a perspective front view of the flexible instrument of FIG. 1.
- (7) FIG. 6 illustrates a partial perspective front view of the surgical access assembly of FIG. 1 with a gear system moving in a first movement.
- (8) FIG. 7 illustrates a partial perspective rear view of the surgical access assembly of FIG. 6.
- (9) FIG. 8 illustrates a perspective view of another surgical access assembly.
- (10) FIG. 9 illustrates a side view of the surgical access assembly of FIG. 8 in a first position.
- (11) FIG. 10 illustrates a side view of the surgical access assembly of FIG. 8 in a second position.
- (12) FIG. 11 illustrates a cut-away view of the surgical access assembly of FIG. 9.
- (13) FIG. 12 illustrates a cut-away view of the surgical access assembly of FIG. 10.
- (14) FIG. 13 illustrates a perspective view of another surgical access assembly.
- (15) FIG. 14 illustrates a top view of the surgical access assembly of FIG. 13.
- (16) FIG. 15 illustrates a side view of the surgical access assembly of FIG. 13.
- (17) FIG. 16 illustrates a perspective bottom view of the surgical access assembly of FIG. 13.

DETAILED DESCRIPTION

(18) Referring now to the discussion that follows and also to the drawings, illustrative approaches to the disclosed assemblies and methods are shown in detail. Although the drawings represent some possible approaches, the drawings are not necessarily to scale and certain features may be exaggerated, removed, or partially sectioned to better illustrate and explain the present disclosure. Further, the descriptions set forth herein are not intended to be exhaustive or otherwise limit or restrict the claims to the precise forms and configurations shown in the drawings and disclosed in the following detailed description.

(19) Described herein is a surgical access assembly, various components for use in same, and a method of using the surgical access assembly. The components disclosed herein provide surgeons with an enhanced ability to minimize trauma to the patient, while providing efficient improved minimally invasive surgical techniques, including imaging of the site. In order to acquire images of the site, ancillary light may be provided to the site. This light may be provided by various light sources within the surgical access assembly. The light sources may also be integrated into a camera assembly and delivered to the surgical site via a tube within a sheath.

(20) More specifically, fluorescent markers may be used to help distinguish certain cells within the body tissue. Certain compounds may cause cancerous tissue to become visible, or illuminate, in response to specific light wavelengths being delivered on to the tissue. This may aid in allowing the surgeon to differentiate and identify tissues. The light sources within the surgical access assembly may deliver this light to the site.

(21) Other types of tissue analysis may be delivered and available via the flexible tool delivery device. In one example, spectroscopic analysis of the tissue via laser interrogation at a specific frequency to interrogate the tissue may be delivered via the flexible tool. In this example, the provided laser may bounce off of the tissue and the receiver of the laser console may classify the tissues. Other forms of tissue interrogation may also be provided via the flexible tool delivery device. Further, biopsy or resection devices, as well as light sources, microscopes, other surgical probes, and tools for intracranial, spinal, abdominal procedures, and more.

(22) The surgical access assembly may include a selectively attachable flexible tool delivery device, having the guide tube of the flexible tool delivery device configured to selectively extend within a lumen of the sheath. The flexible tool delivery device may be relatively small, having a much less diameter than that of the sheath so as to not block other tools or cannulas from being arranged or entering the sheath. Further, the flexible tool delivery device is adjustable in that the position of the distal end of the flexible tool relative to the sheath may be axially adjustable by the flexible tool delivery device. The angle at which the flexible tool delivery device is arranged within the sheath is also adjustable and thus allowing different depths and angles of delivery to the surgical site of the flexible tool. The flexible tool delivery device may attach at the proximal end of the sheath and may be pivotable to adjust the angle of the flexible tool.

(23) The position of the flexible tool may be maneuvered via a single thumb movement. The flexible tool may be a camera, but may also be a biopsy device, grasper, or any flexible device, etc. In the example of the camera, the camera may provide the necessary light to the surgical site to illuminate the tissue.

(24) FIG. 1 illustrates a perspective view of a surgical access assembly **100** having a flexible tool delivery device **104** arranged on a hollow outer sheath **102**. The outer sheath **102** may be defined by a distal end **108** and a proximal end **116** and include a generally hollow body portion **118**. A rim portion **120** is configured as a ring, as illustrated in the drawings. The rim portion **120** may be configured as a grip portion, allowing for engagement by the surgeon. In this example, the rim portion **120** may be referred to as a grip ring and is fixedly secured to body portion **118** at proximal end **116**. The outer periphery may further be provided with a textured surface to provide for ease of gripping the outer sheath **102**. The rim portion **120** may define a plurality of holes **150**.

(25) In one example, the body portion **118** is constructed of a clear biocompatible material that permits viewing of normal tissue, abnormal tissue, as well as critical structures that are disposed outside of body portion **118** when the outer sheath **102** is disposed within such tissue. In one exemplary arrangement, outer sheath **102** is constructed of polycarbonate, though other biocompatible materials may be employed, including resins. The distal end **108** of the outer sheath **102** may include a tapered portion **130**. The tapered portion **130** may serve to ease transitions of the sheath **102** into body tissue.

(26) The outer sheath **102** is not intended to be limited to brain tissue applications, but may be a port configured to receive the flexible instrument. In one other example, the sheath **102** may be used for abdominal procedures. The flexible tool delivery device **104** may be configured to selectively attach to the rim portion **120** of the outer sheath **102**.

(27) The flexible tool delivery device **104** may be any instrument to be flexibly introduced to or proximate to the tissue via the body portion **118** of the outer sheath **102**. In one example, the flexible tool delivery device **104** may be configured to deliver a camera with a filter configured to pick up fluorescent images generated by the tissue. The flexible tool delivery device **104** may include a guide tube **140** extending from a hub **142** into the outer sheath **102**, without substantially

filling or blocking the lumen of the sheath **102**. The guide tube **140** may extend at least partially down three of the sheath **102** within the lumen. The guide tube **140** may be rigid enough to form a generally straight guide within the sheath **102**. This is discussed in more detail below.

(28) The flexible tool delivery device **104** includes the hub **142** configured to rest at the rim portion **120**. The hub **142** may selectively clamp onto the rim portion **120** via a plurality of mechanisms. In the example shown, the hub **142** may include at least one pin **144**. The at least one pin **144** may include a pair of pins arranged on each side of a body **148** of the hub **142**. The holes **150** in the rim portion **120** may be configured to receive the at least one pin **144** and secure the flexible tool delivery device **104** to the rim portion **120** of the sheath. In the example shown, the holes **150** are equally spaced about the rim portion **120**, thus allowing the pins **144** to be received at any two consecutive holes **150** around the rim portion **120**. Other mechanisms for attaching the hub **142** to the sheath **102** may also be contemplated.

(29) As best seen in FIGS. **1**, **2**, **3**, **5**, and **7**, the hub **142** of the flexible tool delivery device **104** may define an aperture **152** configured to receive a tool **138**. FIG. **2** illustrates a partial side view of the surgical access assembly **100** of FIG. **1** with the tool **138** in an uninstalled position where the flexible tool **138** has not yet been inserted into the aperture **152**. FIG. **3**, on the other hand, illustrates a partial side view of the surgical access assembly of FIG. **1** with the flexible tool **138** in an installed position. As explained above, the tool **138** may include any number of surgical tools including but not limited to a camera assembly, laser, fiberoptic device, resection device, biopsy device, wand, suction device, etc.

(30) The tool **138** may be inserted into aperture **152** and received by a channel **154** defined by the hub **142**. The channel **154** may be curved so as to guide the tool **138** downward into the guide tube **140**. For this purpose, the tool **138** may be made of a bendable or malleable material such that the tool **138** may bend to conform to the curve of the channel **154**, but also maintain a relatively straight form when extending through the guide tube **140** and into the sheath **102**.

(31) Once inserted into the aperture **152**, the tool **138** may be selectively driven into the lumen of the outer sheath **102**. In the example shown, the tool **138** may be fed through the aperture by a gear set **158**. The channel **154** guide the tool **138** into the guide tube **140**. The gear set **158** may move the tool **138** into the guide tube **140**, towards the distal end **108** of the sheath **102** or retract the tool **138** away from the distal end **108**.

(32) Referring to the figures, including FIGS. **4** and **5**, the gear set **158** may include a pair of gears **160** configured to interlock with one another and rotate in opposite directions. That is, as a first gear **160a** rotates clockwise, the other second gear **160b** rotates in a reciprocal direction, or counterclockwise. The gear set **158** may be controlled by a control knob **162**. The control knob **162** may be arranged at a top or upper side of the hub **142** and may be configured to selectively rotate the gears of the gear set **158**. A drive wheel **164** may be coupled to the control knob **162** as well as the first gear **160a**. When the drive wheel **164** rotates in a first direction, so does the drive wheel **164** and the first gear **160a**.

(33) Similarly, a follow wheel **166** may be coupled to the second gear **160b** and configured to rotate with the second gear **160b**. the second gear **160b** may rotate upon rotation of the first gear **160a**. Thus, when the first gear **160a** and the drive wheel **164** are rotated by the control knob **162**, the second gear **160b**, as well as the follow wheel **166**, are also rotated. Thus, the drive wheel **164** and follow wheel **166** are both driven the by the control knob **162**.

(34) The drive wheel **164** and follow wheel **166** may be cylinders. The drive wheel **164** and the follow wheel **166** are fixed on either side of the channel **154** and are configured to engage with the tool **138**. That is, the drive wheel **164** and follow when **166** may “pinch” the tool **138**.

Consequently, when the drive wheel **164** and follow wheel **164** rotate, the tool **138** may be translated in that direction. The drive wheel **164** and follow wheel **166** may also maintain or hold the tool **138** at a fixed position once a desired location of the tool **138** has been reached. The drive wheel **164** and the follow wheel **166** may be made of silicone or a similar material having some

compliance in order to accommodate varying diameters of the flexible tool **138**.

(35) Referring specifically to FIGS. **6** and **7**, where FIG. **6** illustrates a partial perspective front view and FIG. **7** illustrates a partial perspective rear view of the surgical access assembly **100** with the gear system **158** moving in a first movement. In this example, a counterclockwise rotation of the control knob **162** may cause gears to rotate towards one another and the tool **110** to move into the guide tube **140** within sheath **102** towards the distal end thereof.

(36) A counterclockwise motion, or second movement, may reverse the direction of the gears **160**, causing each to rotate away from each other, causing tool **138** to retract and move away from the distal end **108** of the sheath **102**. Accordingly, by rotating the control knob **162**, the length of the position of the tool **138** relative to the sheath **102** may be adjusted. The control knob **162** may include projections and define ridges configured to allow for easy actuation or rotation by the user. In one example, a surgeon's thumb may rotate the knob **162** in a clockwise or counterclockwise direction so as to adjust the position of the distal end of the flexible tool **138** within the sheath **102**. Because the hub **142** is arranged on a side of the sheath **102** as opposed to be arranged at the center or above the sheath **102**, visibility down the sheath **102** is maintained.

(37) As explained, an instrument or flexible tool **138** may be delivered through the guide tube **140**. The instrument may include the camera and/or a light source to acquire images of the surgical site to permit visualization of tumors, vessels, fiber tracks, fascicles and even healthy tissue, in real-time. The camera may enable the surgeon to have real-time information about the fiber tract and fascicle location, after placement of outer sheath **102**. The camera may also provide light at specific wavelengths to illuminate fluorescents within the tissue at the surgical site. In one example, the camera may be housed within the hub **142**. The camera may be a fiberoptic camera having glass fibers configured to deliver light. The camera may be a CMOS camera located on the distal end of the tool **138**. In other examples, the camera arranged within the hub. The camera could also be independent of the hub **142** and guide tube **140**.

(38) In addition to the length of the flexible tool **138** extending from the guide tube **140** within the sheath **102** being adjustable, the angle of the flexible tool **138** may also be adjustable. This may be achieved by pivoting the hub **142** at the rim portion **120** in order to adjust the relative angle of the tube **140** extending therefrom. This may allow the surgeon to easily adjust the light source and/or camera without the need to reposition the outer sheath **102**, avoiding unnecessary intrusion and trauma to the nearby tissue.

(39) While the guide tube **140** is discussed herein, the hub **142** itself may deliver the tool to the proximal end of the sheath without the use of the guide tube **140**. For example, the hub **142** may be configured to receive a camera and angle the camera down the sheath **102** to image the tissue proximal to the distal end of the sheath **102**. Thus, the flexible tool may be delivered to the tissue.

(40) FIG. **8** illustrates a perspective view of another surgical access assembly **200**. This example assembly **200** may include a sheath **202** and a hub **242** maintaining a guide tube **240**. Depth markings may be included to aid in identifying the relative position of the tube **240** within the sheath **202**.

(41) FIG. **9** illustrates a side view of the surgical access assembly **200** of FIG. **8** in a first position while FIG. **10** illustrates a side view of the surgical access assembly **200** of FIG. **8** in a second position. FIG. **11** illustrates a cut-away view of the surgical access assembly **200** of FIG. **9** and FIG. **12** illustrates a cut-away view of the surgical access assembly **200** of FIG. **10**.

(42) FIG. **13** illustrates a perspective view of another surgical access assembly **300** having a flexible instrument **304** arranged on a hollow outer sheath **302**. A rim portion **320** is configured as a ring, as illustrated in the drawings. The rim portion **320** may be configured as a grip portion, allowing for engagement by the surgeon as well as configured to receive a hub **342** of the flexible instrument **304**, similar to the example shown in FIG. **1**. As explained, the flexible instrument **304** may include a guide tube **140** extending down into the outer sheath, without substantially filling or blocking the lumen of the sheath **102**.

(43) Referring also to FIGS. 14, 15 and 16, the flexible instrument 304 includes the hub 342 configured to rest at the rim portion 320. The hub 142 may selectively secure onto the rim portion 320 via a plurality of mechanisms. In the example shown, the hub 342 may include at least one projection 344, as best illustrated in FIG. 15. The at least one projection 344 may include a pair of projections arranged on each side of a body of the hub 342. A plurality of equally spaced holes 350 defined along the rim portion 320 may be configured to receive the at least one projection 344 and secure the flexible instrument 304 to the rim portion 320 of the sheath 302. As explained, in the example shown, the holes 350 are equally spaced about the rim portion 320, thus allowing the pins 144 to be received at any two consecutive holes 150 around the rim portion 120. Other mechanisms for attaching the hub 142 to the sheath 102 may also be contemplated.

(44) The projections 344 may include a distal end 380 configured to be snap-fit through the holes 350. The distal end 380 may include a malleable pair of cones configured to be forced together upon insertion into the hole 350, and then biased outwardly back to their resting position once the projection 344 has proceeding through the hole 350 to snap fit the projection within the hole 350. The distal end 380 may, additionally or alternatively, include a bolt, or other securing mechanism, to aid in maintaining the projections 344 within the holes 350.

(45) It will be appreciated that the surgical access system and methods described herein have broad applications. The foregoing embodiments were chosen and described in order to illustrate principles of the methods and apparatuses as well as some practical applications. The preceding description enables others skilled in the art to utilize methods and apparatuses in various embodiments and with various modifications as are suited to the particular use contemplated. In accordance with the provisions of the patent statutes, the principles and modes of operation of this disclosure have been explained and illustrated in exemplary embodiments.

(46) It is intended that the scope of the present methods and apparatuses be defined by the following claims. However, it must be understood that this disclosure may be practiced otherwise than is specifically explained and illustrated without departing from its spirit or scope. It should be understood by those skilled in the art that various alternatives to the embodiments described herein may be employed in practicing the claims without departing from the spirit and scope as defined in the following claims. The scope of the disclosure should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the arts discussed herein, and that the disclosed systems and methods will be incorporated into such future examples. Furthermore, all terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. It is intended that the following claims define the scope of the invention and that the method and apparatus within the scope of these claims and their equivalents be covered thereby. In sum, it should be understood that the invention is capable of modification and variation and is limited only by the following claims.

Claims

1. A surgical access assembly, comprising an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween; and a flexible tool delivery device having a hub and a guide tube, the hub configured to selectively secure to a rim at the proximal end of the outer sheath, and the guide tube configured to extend at least partially into the outer sheath and provide a flexible tool proximate to a tissue, wherein the hub includes at least one pin and wherein the rim defines at least one hole configured to receive the pin of the hub to secure

the hub to the sheath.

2. The assembly of claim 1, wherein the hub is configured to receive the flexible tool and provide the flexible tool to the guide tube.

3. The assembly of claim 2, wherein the hub includes a pair of wheels configured to receive the flexible tool therebetween and drive the flexible into and out of the guide tube.

4. The assembly of claim 3, wherein the hub includes at least one control knob configured to selectively rotate the wheels such that the flexible tool is frictionally engaged between the wheels to as to be guided into the guide tube towards the distal end of the guide tube when the control knob is rotated in a first direction, and where the flexible tool is retracted in the opposite direction towards a proximal end of the guide tube when the control knob is rotated in an opposite second direction.

5. The assembly of claim 3, wherein the wheels are mailable and configured to receive flexible tools with varying diameters.

6. The assembly of claim 3, wherein the hub includes a channel connected to the guide tube and configured to receive the flexible tool, the wheels are arranged on each side of the channel such that the flexible tool is movable within the channel and guide tube.

7. The assembly of claim 1, wherein the hub is pivotable at the rim of the sheath such that the angle of the guide tube is selectively adjustable based on the relative angle of the hub with respect to the sheath.

8. The assembly of claim 1, wherein the pin includes a projection at a distal end of the pin configured to snap fit within the hole.

9. The assembly of claim 1, wherein the hub includes at least two pins and wherein the rim defines a plurality of equally spaced holes configured to receive the pins of the hub to secure the hub to the sheath.

10. A flexible tool delivery device for a surgical system, comprising a hub configured to selectively secure to a rim at a proximal end of an outer sheath of the surgical system, and a guide tube coupled to the hub and extending at least partially into the outer sheath to provide a flexible tool proximate to a tissue during use, wherein the hub includes at least one pin and wherein the rim defines at least one hole configured to receive the pin of the hub to secure the hub to the sheath.

11. The flexible tool delivery device of claim 10, wherein the hub is configured to receive the flexible tool and provide the flexible tool to the guide tube.

12. The flexible tool delivery device of claim 11, wherein the hub includes a pair of wheels configured to receive the flexible tool therebetween and drive the flexible tool into and out of the guide tube.

13. A flexible tool delivery device for a surgical system, comprising a hub configured to selectively secure to a rim at a proximal end of an outer sheath of the surgical system, and a guide tube coupled to the hub and extending at least partially into the outer sheath to provide a flexible tool proximate to a tissue during use, wherein the hub is configured to receive the flexible tool and provide the flexible tool to the guide tube, wherein the hub includes a pair of wheels configured to receive the flexible tool therebetween and drive the flexible tool into and out of the guide tube, and wherein the hub includes at least one control knob configured to selectively rotate the wheels such that the flexible tool is frictionally engaged between the wheels to as to be guided into the guide tube towards the distal end of the guide tube when the control knob is rotated in a first direction, and where the flexible tool is retracted in the opposite direction towards a proximal end of the guide tube when the control knob is rotated in an opposite second direction.

14. The flexible tool delivery device of claim 12, wherein the wheels are mailable and configured to receive flexible tools with varying diameters.

15. The flexible tool delivery device of claim 12, wherein the hub includes a channel connected to the guide tube and configured to receive the flexible tool, the wheels are arranged on each side of the channel such that the flexible tool is movable within the channel and guide tube.

16. The flexible tool delivery device of claim 10, wherein the hub is pivotable at the rim of the sheath such that the angle of the guide tube is selectively adjustable based on the relative angle of the hub with respect to the sheath.
17. A surgical access assembly, comprising an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween; and a flexible tool delivery device having a hub configured to selectively secure to a rim at the proximal end of the outer sheath and selectively receive a flexible tool to deliver the flexible tool, proximate to a tissue at a distal end of the outer sheath, wherein the hub includes at least one pin and wherein the rim defines at least one hole configured to receive the pin of the hub to secure the hub to the sheath.
18. A surgical access assembly, comprising an outer sheath defined by an open distal end and an open proximal end and including a hollow body portion therebetween; and a flexible tool delivery device having a hub and a guide tube, the hub configured to selectively secure to a rim at the proximal end of the outer sheath, and the guide tube configured to extend at least partially into the outer sheath and provide a flexible tool proximate to a tissue, wherein the hub is configured to receive the flexible tool and provide the flexible tool to the guide tube, wherein the hub includes a pair of wheels configured to receive the flexible tool therebetween and drive the flexible into and out of the guide tube, and wherein the hub includes at least one control knob configured to selectively rotate the wheels such that the flexible tool is frictionally engaged between the wheels to as to be guided into the guide tube towards the distal end of the guide tube when the control knob is rotated in a first direction, and where the flexible tool is retracted in the opposite direction towards a proximal end of the guide tube when the control knob is rotated in an opposite second direction.
19. The assembly of claim 18, wherein the hub includes a gear assembly coupled to the control knob and configured to rotate the wheels in response to rotation of the control knob.
20. The flexible tool delivery device of claim 13, wherein the hub includes a gear assembly coupled to the control knob and configured to rotate the wheels in response to rotation of the control knob.
21. The flexible tool delivery device of claim 12, wherein the hub includes at least one control knob configured to selectively rotate the wheels such that the flexible tool is frictionally engaged between the wheels to as to be guided into the guide tube towards the distal end of the guide tube when the control knob is rotated in a first direction, and where the flexible tool is retracted in the opposite direction towards a proximal end of the guide tube when the control knob is rotated in an opposite second direction.
22. A flexible tool delivery device for a surgical system, comprising a hub configured to selectively secure to a rim at a proximal end of an outer sheath of the surgical system, and a guide tube coupled to the hub and extending at least partially into the outer sheath to provide a flexible tool proximate to a tissue during use, wherein the hub is configured to receive the flexible tool and provide the flexible tool to the guide tube, wherein the hub includes a pair of wheels configured to receive the flexible tool therebetween and drive the flexible into and out of the guide tube, and wherein the hub includes a channel connected to the guide tube and configured to receive the flexible tool, the wheels are arranged on each side of the channel such that the flexible tool is movable within the channel and guide tube.
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